Regular Articles

Ugochukwu Nwachukwu, Daniel O. Aikhuele and Shahryar Sorooshian*

Dynamic concept drift detection using an integrated machine learning model with error-based Page-Hinkley Test for enhanced predictive accuracy in adhesive manufacturing process

https://doi.org/10.1515/eng-2025-0158
Received February 13, 2025; accepted November 23, 2025;
published online January 23, 2026

Abstract: In response to the need for dynamic adaptability in adhesive manufacturing, this paper introduces a machine learning model and a new error-based concept drift detection mechanism based on Page-Hinkley Test (PHT) which is a statistical method used for change point detection in time-series data for enhancing accuracy and robustness in adhesive manufacturing process. Upon its implementation, the model monitors the prediction error in the manufacturing process such that it dynamically adjusts its responses in the presence of significant deviation. The integrated machine learning and PHT-based approach successfully detected multiple drift points, starting at t = 1,007 and going up to t = 1,342, which covers both abrupt and gradual shifts. Contrasting with the model, the study was able to work with only one pre-defined drift point, such as t = 1,000, after each detection of drift, the cumulative error resets, allowing the model to relearn from the present data distribution. This repeated recalibration yielded a final root mean square error (RMSE) of 2.0550, showing a great error reduction, thereby improving the product quality. This increased flexibility is very useful for adhesives production within the oil and gas industry, where variation in material quality, environmental conditions, or equipment performance might otherwise affect product quality. Compared with the traditional static model, the integrated machine learning and PHT-based system greatly improved the accuracy and response time of predictions, demonstrating its value in real-time quality control and defect rate minimization for manufacturing.

Keywords: adhesive manufacturing; Page-Hinkley Test; error-based concept drift detection; adaptive machine learning; variability

1 Introduction

The modern manufacturing process is dynamic in nature, and it requires adaptive and resilient systems that would respond to such variations with a minimum loss of quality and performance. Such variations are particularly critical in sectors like the adhesive materials production, where production parameters are so different due to raw material properties, environmental conditions, and processing parameters. Among all the materials fabrication processes, adhesive materials are some of the most challenging, as they rely on specific formulation and conditions that must be precisely controlled to achieve consistent quality, performance, and reliability in the final product. However, these conditions of variability and complexity are often difficult to apply to traditional process control models, which may be inefficient and possibly inconsistent in their products.

The manufacturing conditions remain one of the most influential variables that ensure quality and consistency in adhesive materials. Because controlled processes are generally applied during their formulation, the adhesive properties usually remain stable. Various studies into adhesive manufacturing therefore exhibit investigations on the influences of temperature, humidity, and process optimization on product variability. Besides considering novel adhesive formulations, insights on these will serve to increase the demand for eco-friendly and efficient manufacturing in turn. Landgraf et al. [1] performed the simulation of adhesive curing processes. It can be seen from their work that, for reaching a homogeneous curing in adhesive layers, careful temperature

***Corresponding author: Shahryar Sorooshian**, Department of Business Administration, University of Gothenburg, Gothenburg, Sweden; and Faculty of Engineering and Sustainable Development, University of Gävle, Gävle, Sweden E-mail: shahryar.sorooshian@gu.se. https://orcid.org/0000-0001-5336-827X
Ugochukwu Nwachukwu and Daniel O. Aikhuele, Department of Mechanical Engineering, University of Port Harcourt, East-West Road, Choba-Port Harcourt, Rivers State, Nigeria. https://orcid.org/0000-0001-9274-4530 (D.O. Aikhuele)

Open Access. © 2026 the author(s), published by De Gruyter. This work is licensed under the Creative Commons Attribution 4.0 International License.

control and material stability are necessary in manufacturing where composite structures are bonded together in highly stressed applications such as aerospace or electronics manufacturing. Their findings showed that curing variability can result in reduced adhesion strength and the failure of products, determining process consistency as one of the most relevant factors to ensure adhesive manufacturing reliability. Song and Kim [2], researched on polyurethane adhesive delamination in Cross Laminated Timber (CLT) due to environmental conditions. They were able to show, by altering temperature and humidity conditions, that during winter months when manufacturing occurs, there will be more instances of adhesion failure, hence explaining how seasonal changes result in degraded performance of adhesives. This again emphasizes the process adaptations that have to be done in the adhesive manufacturing processes in order to stabilize the product quality under variable ambient conditions.

The inability in adhesive properties does drive innovation in eco-friendly adhesive materials. Zheng et al. [3] developed a soybean flour-sugar solution adhesive with the aim of optimizing the pressing parameters for maximum bond strength in plywood production. Their work thus illustrates a potential of bio-based adhesives as viable alternatives, though their performance remains sensitive to various conditions of technological processing, including temperature and curing time. The current study aligns with broader industry trends focused on balancing performance with environmental sustainability. Islam et al. [4] studied the bio-based adhesives from natural rubber latex blended with starch for the production of particleboards. Their work determined the conditions of pressing that provide optimum strength and stability of the adhesive. This, together with many other developments, has shown that though the production parameters for bio-based adhesives need to be stringently controlled, they can meet the required performance parameters and present feasible options in the adhesive-manufacturing industries.

In the high-tech sector, Moss et al. [5] presented advanced automated adhesive-dispensing systems used in precision electronics; controls were instituted to ensure uniform adhesive application in micro-manufacturing. This automation indicates a line of development to reduce variation by technological innovation since consistent adhesive application is critical for component durability in manufacturing electronics and therefore subject to circuit failures due to variations. Cumulatively, these studies have shown that understanding and learning to manage the variability in the production of adhesives is important. Improved process controls in an adaptive manufacturing environment enable manufacturers to allow more natural variability that improves quality while maintaining product sustainability.

Also, several studies have proven, that the environmental conditions of temperature and humidity are one of the most relevant variables when dealing with bonded joints. In fact, according to Machado et al. [6], small variations in environmental factors give rise to significant changes in mechanical properties under impact loadings. For example, the automotive and defense industries depend on adhesive joints that should withstand collision and ballistic impact; thus, the need for standard test methods that takes such factors into consideration is immediate. According to Viana et al. [7], the improvement of adhesive applications under extreme stress requires an understanding of the interactions coming from the coupling of environmental conditions with the mechanical loads. Another important factor affecting variability, and falling within, is the formulation of adhesive materials. Taking, for instance, the case study of the usage of grinded beech bark as filler in mixtures of urea formaldehyde adhesive. It will be observed that changes in fillers would greatly alter values such as thermal conductivity, viscosity, and ultimately affect the adhesives' performance [8]. The filler composition's variability emphasizes the priority for eco-friendly materials in adhesive formulations and further improving performance in a way not harmful to the environment.

The mechanical behaviour will also be different across adhesives, as particular formulations may well be made, as Ribas et al. [9] pointed out. They conducted a study of the strain rate-dependent properties, finding that adhesive performance could be influenced when different curing times and temperatures were used under certain loading modes. This finding has further burdened manufacturers to optimize processing conditions in order to improve adhesive reliability, especially in critical applications. During the manufacturing of composites, the idea of tack in prepregs is very important, and minor deviations in manufacturing conditions are going to alter tack properties considerably in such cases [10]. This would likely involve such complex interplay between the adhesive and cohesive phenomena that stringent control of machine parameters and environmental conditions would be required to avoid defects and ensure stability of the process. Understanding such interactions is of paramount importance to any manufacturer concerned with the production of high-quality composite materials.

Although the progress to understand the production variability within adhesives has been made, a number of knowledge gaps still exist. First, while several studies have demonstrated environmental conditions and filler compositions, recommended practices for testing the performance of adhesives under varied conditions are still not proposed. The further work that should be undertaken may focus on

the development of guidelines for comprehensive testing of adhesives, taking into account a wide range of environmental variables. Moreover, the detailed behavior of new eco-compatible fillers requires additional discussion. Long-term effects of these fillers on the performance of adhesives in service might be very important in those cases where industries are moving towards sustainability. The last gap that should receive considerable attention concerns is the interaction of manufacturing processes with the long-term degradation processes of adhesives as a function of environmental exposure. Understanding how these factors contribute to adhesive failure mechanisms could lead to the development of more resistant adhesive systems.

Machine learning has become a strong approach in recent years for predictive maintenance, quality assurance, and process optimization during manufacturing [11]. These models analyze vast volumes of data, recognize patterns, and make data-driven decisions in real time with a view to improving production efficiency and output quality. Although machine learning bears a promising potential, developing models that can adapt to a continuously changing production environment remains a challenge. In fact, manufacturing conditions often change due to inconsistency in raw materials, wear and tear of machines, and changes in environmental factors, among others; hence, an adaptive model will be required that can dynamically recalculate and learn from new data [12].

This study is concerned with the development of an adaptive machine learning model for adhesive manufacturing to address real-time variability, with special consideration given to operational performance optimization and product quality. The approach adopted herein differs from the previous static model in that it adopts an adaptive framework, updating the parameters based on recent data, responding to changes in manufacturing conditions, such as temperature variations, changes in the viscosity of the raw materials, and adjustment to processing time. Applying this adaptive model will enhance the robustness of manufacturing against uncertainty and reduce waste, hence improving consistency in adhesive properties. The novelty of this study therefore lies in the error-driven adaptive calibration of the Page–Hinkley Test (PHT) threshold and its integration with a forgetting factor (λ) for dynamic reweighting of recent data. This hybrid mechanism enables faster adaptation to process drift and improved stability in real-time learning. In addition, its application to adhesive material manufacturing represents a new domain-specific implementation of PHT under complex, time-varying production conditions. This sets important implications on precision-driven industries that require adaptability.

2 Development of adaptive machine learning model

In this section, a system to dynamically respond to real-world changing conditions with little intervention from humans has been developed with the backend parameters presented. This system is based on adaptive machine learning model for real-time, continuous learning in manufacturing environments and it is aimed at addressing critical and dynamic issues relating to product quality in the manufacturing industry.

The selection of the hyperparameters, i.e., learning rate (η = 0.01), forgetting factor (λ = 0.99), and regularization coefficient (α = 0.001), are motivated by the empirical stabilization and convergence analyses of adaptive process modeling and real-time controller structures applied to manufacturing processes [13, 14]. These parameters guarantee that the model is sensitive to the process drifts and is resistant to the temporary changes in sensor values or process noise.

(i) Data Acquisition

It is important to note that the study adopted a synthetic dataset which is predetermined by controlled experimentation, transparency, and reproducibility of the adaptive drift detection framework assessment. Real production datasets in adhesive manufacturing with well-labeled concept drifts are scarce as proprietary issues and continuous sensor-based measures in factories are expensive to implement. In turn, the synthetic dataset offers a managed and transparent setup to study the model behavior to known drift points based on realistic process dynamics based on adhesive production systems.

The dataset was also structured in a manner that it represented the fundamental nature of adhesive manufacturing processes, where temperature and pressure are the key drivers of the process that affect the viscosity, curing rate, and tack strength. Previous investigations have indicated that variations in curing temperature and pressure profiles have dramatic influence on the rheology of adhesives resulting in differences in cross-linking kinetics and bond strength. In this regard, the pre-drift and post-drift parameter settings in this research approximate changes in operation between optimal curing conditions and degenerated conditions due to raw material variation, equipment wear, or environmental variability. Gaussian noise (σ = 0.5) is added to mimic sensor error and environmental interference, and relates to the stochastic real-world characteristics of the process of a sensor-based industrial acquisition system.

(ii) Mathematical Formulation of the Model

In an online learning scenario, data arrives sequentially as (x_t, y_t) for $t = 1, 2, \ldots, T$, where $x_t \epsilon \mathbb{R}^d$ is the feature vector. $y_t \epsilon \mathbb{R}$ (regression) or $y_t \epsilon \{1, \ldots, C\}$ (classification) on the other hand is the target variable. The goal here however is to learn a predictive model $f_t(x)$ at each time step (t) that minimizes the cumulative loss, and it is given as [15];

$$\min \sum_{t=1}^{T} \mathscr{L}(f_t(x), y_t) \tag{1}$$

where $\mathscr{L}$ is a loss function, such as Mean Squared Error (MSE) for regression or Cross-Entropy Loss for classification.

If the model is assumed to be a linear model for simplicity, therefore it can be rewritten as;

$$f_t(x) = W_t^T x + b_t \tag{2}$$

where W_t are the weights and b_t is the bias at time (t) for the model. The update rule for the weights used in the model can be derived using Stochastic Gradient Descent (SGD) which is given as;

$$W_{t+1} = W_t - \eta_t \nabla_W \mathscr{L}(f_t(x), y_t) \tag{3}$$

$$b_{t+1} = b_t - \eta_t \frac{\delta \mathscr{L}}{\delta b} \tag{4}$$

where η_t is the learning rate at time (t), for the MSE loss, the equation is given as shown in equation (5) while the gradients are given in equations (6) and (7) finally, the equations in which the system rely on for its update is given equations (8) and (9). They represent the baseline stochastic gradient descent (SGD) updates, while Equations (10) and (11) extend these updates by introducing the forgetting factor (λ), enabling adaptive weighting of recent observations under concept drift.

$$\mathscr{L}(f_t(x), y_t) = \frac{1}{2}(y_t - f_t(x))^2 \tag{5}$$

$$\nabla_W \mathscr{L} = -(y_t - W_t^T x_t + b_t) x_t \tag{6}$$

$$\frac{\delta \mathscr{L}}{\delta b} = -(y_t - W_t^T x_t + b_t) \tag{7}$$

$$W_{t+1} = W_t - \eta_t (y_t - W_t^T x_t + b_t) \tag{8}$$

$$b_{t+1} = b_t - \eta_t (y_t - W_t^T x_t + b_t) \tag{9}$$

When the system is used for handing concept drift, the underlying data distribution changes over time. Manufacturing environments is a typical example that is prone to such concept drift, however, to enable the system adaptation, this study has incorporated a function termed the forgetting factor and it is given as $\lambda \epsilon (0, 1)$. It assigns higher weight to recent data while gradually reducing the influence of outdated information. This mechanism is crucial for addressing time-dependent variations and concept drift in dynamic environments. Accordingly, to capture the gradual variations in the manufacturing data, the model updates its parameters using an adaptive online learning scheme expressed as:

$$W_{t+1} = \lambda W_t - \eta_t ((y_t - W_t^T x_t) + b_t) \tag{10}$$

$$b_{t+1} = \lambda b_t - \eta_t ((y_t - W_t^T x_t) + b_t) \tag{11}$$

The adaptive learning rate η_t governs the step size for weight updates and is modulated based on the recent magnitude of prediction errors to prevent instability after drift events. Similarly, methods like Windowed Learning or Adaptive Regularization can also be employed. However, to ensure the model generalization, especially in noisy like the manufacturing environments, the regularization term $R(W)$ is added to the loss function.

$$\mathscr{L}_{\text{Reg}} = \mathscr{L} + \alpha R(W) \tag{12}$$

Common choices of the regularization include, L2 Regularization which is given as $R(W) = \frac{1}{2}\|W\|^2$ and the L1 Regularization $R(W) = \frac{1}{2}\|W\|_1$. With this however, the gradient updates can be rewritten as;

$$\nabla_W \mathscr{L}_{\text{Reg}} = \nabla_W \mathscr{L} + \alpha \nabla_W R(W) \tag{13}$$

Upon detecting a drift, the system adapts the model by updating the parameters through a continue online updates with new data and through an ensemble learning to maintain multiple models and weigh them based on their performance in current conditions. While the linear model above is very straightforward, a more complex model like Neural Networks can be employ to capture intricate patterns in the manufacturing data.

The adaptation of machine learning models for real-time manufacturing can be achieved by blending online learning algorithms with efficient adaptation strategies and automation to keep human intervention at a bare minimum [16]. Considering the nature of incremental updates, concept drift handling, and automated hyperparameter tuning, such models would easily set in or adapt to rapid changes in manufacturing conditions without degradation in performance. This would create a fundamental mathematical backbone through which such adaptive systems could be lucratively implemented.

(iii) Page-Hinkley Test (PHT)

To ensure, further development of the above model into the adaptive learning system at the level of setting observation and study of signal or process for detecting abrupt changes, a testing module called the Page-Hinkley Test (PHT) has been proposed. The PHT is a statistical test used when trying to

detect shifts in time series data [17]. It captures the trend of a sequentially collected data set and is therefore useful in detecting changes, drifts or anomalies in processes and serves a quality control tool. Its application in the paper, is mainly to avoid delay in determining when distribution changes will occur and to avoid false alarms. PHT maintains a running sum of the deviations of the signal from the mean; an alarm is triggered whenever the sum exceeds a threshold that is pre-set. The approach works primarily in four steps:

(a) Calculate Cumulative Mean: The mean of the observed data is tracked over time.
(b) Deviation from Mean: For every new data arrival, the deviation from the cumulative mean is computed.
(c) Cumulative Deviation: These deviations are then summed up in a running total.
(d) Thresholding: The test signals a probable change if this cumulative deviation exceeds some threshold.

Compared to other drift detectors approach, the PHT offers several advantages, for example, the CUSUM (Cumulative Sum) test is very sensitive to minor, continuous changes in mean but can have high levels of false alarms in non-stationary or noisy conditions. By comparison, PHT is more resistant to noise and can better detect both sharp and smooth shifts in a distribution, so is more reasonable in time-sensitive industrial data. The DDM (Drift Detection Method) is based on the measurement of the error rate of a classifier and can only work when labeled samples are continuously available, which is not always the case in a manufacturing environment. The ADWIN (Adaptive Windowing) technique adapts the size of the window dynamically to identify the presence of drifts but may be computationally costly with high-frequency data streams. On the same note, the EWMA (Exponentially weighted moving average) method eliminates short term fluctuations but has the disadvantage of lagging behind regarding the detection of sudden changes, since it involves averaging.

Therefore, the Page-Hinkley Test is adopted in this study because it provides an efficient balance between sensitivity, computational simplicity as well as robustness. It adapts effectively to non-stationary environments where it requires minimal parameter tuning, and performs well in scenarios involving both abrupt and gradual drifts – making it highly suitable for real-time adaptive learning and predictive maintenance applications.

(iv) Simulation / Experimental Setup

In this regard, the pre-drift and post-drift parameter settings in this research approximate changes in operation between optimal curing conditions and degenerated conditions due to raw material variation, equipment wear, or environmental variability. Gaussian noise (σ = 0.5) is added to mimic sensor error and environmental interference, and relates to the stochastic real-world characteristics of the process of a sensor-based industrial acquisition system. This is carried out by following these steps:

(i) Preprocessing: Cleaning, normalization, and feature extraction of the data.
(ii) Model Selection: The selection of a suitable model that supports online learning.
(iii) Continuous Learning Mechanism: Incremental model update algorithms that arrive successively with new data.
(iv) Adaptation Strategy: Methods for detecting the occurrence of concept drifts or sudden changes and adapting to them.
(v) Automation Pipeline: Integrating all parts into one whole in order to minimize manual intervention.

For the model performance, the root mean squared error (RMSE) for regression predictions was calculated and this followed by a sensitivity analysis was conducted to evaluate the influence of the forgetting factor (λ) and learning rate (η) on model performance. Values of λ between 0.95 and 0.995 maintained stability while enabling effective adaptation to drift, whereas ηvalues beyond 0.02 caused oscillations in convergence. The selected λ = 0.99 and η = 0.01 thus represent a stable trade-off between responsiveness and robustness. The algorithmic pseudocode flowchart for the adaptive online learning with concept drift detection has been presented In Figure 1.

3 Results of the implementation of an adaptive machine learning model for changing manufacturing conditions in an adhesive material manufacturing company

The problem addressed in this section is the need for an adaptive machine learning model that can learn from real-time data and adapt to the changing manufacturing conditions in an adhesive material company serving the oil and gas industry in Nigeria. The specific challenges include;

(i) Dynamic manufacturing environment with changing production conditions.
(ii) Need for continuous learning from real-time data with minimal human interference.
(iii) Sudden changes in production conditions (concept drift) affecting product quality.

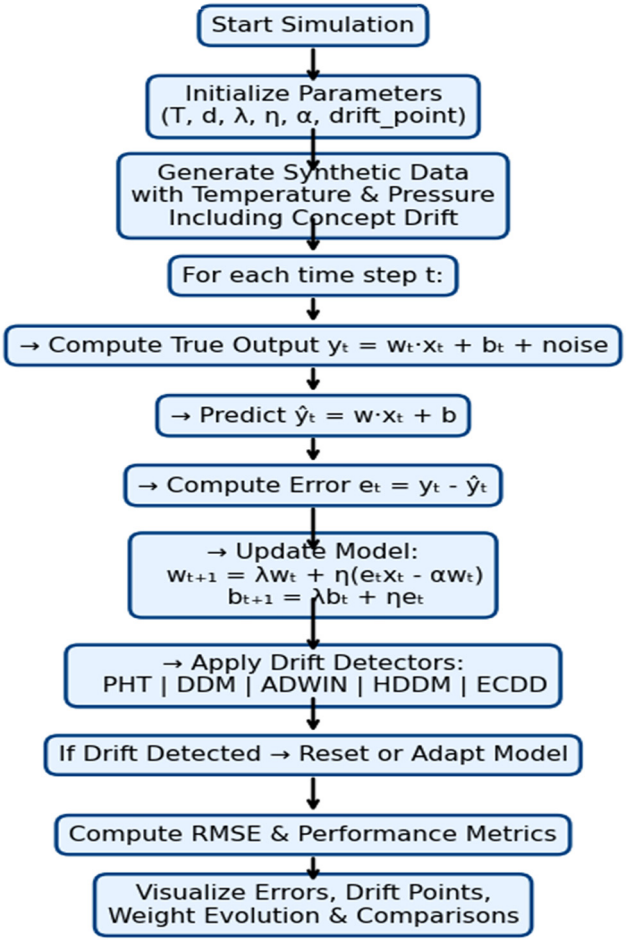


Figure 1: The algorithmic pseudocode flowchart for the model.

Hence, to address these issues, an online learning model was developed and validated for the concept drift, here the model uses a forgetting factor (λ = 0.99) to balance past and recent data, a learning rate (η = 0.01) for steady updates, and L2 regularization (α = 0.001) to prevent overfitting.

In order to reflect the dynamic characteristics of the manufacturing process, a synthesized data set with 2000 samples were used, having two prominent features (temperature and pressure) which have co-efficient which are referred to as Weight 1 and Weight 2 respectively and are used in predicting the quality of the adhesive material. Concept drift was induced at t = 1,000 by changing the true weights from [2.0, −3.5] and bias 5.0 to [−1.0, 4.0] and bias −2.0, reflecting abrupt shifts in manufacturing conditions such as changes in raw materials or equipment performance.

Results from the simulation as presented in Table 1, shows that the model effectively demonstrated adaptation to both stable and drifted conditions, achieving a final RMSE of 3.1351. However, prior to the drift, the predicted values can be seen to be closely matched with the true values, indicating accurate learning in a stable production phase. After the drift on the other hand, the model dynamically adjusted its weights and bias in response to the detected changes, maintaining predictive accuracy and ensuring timely responses to deviations in the product quality. This continuous adaptation is very critical for sustaining the produced adhesive quality despite the fluctuations in the temperature as well as pressure and the raw material properties.

Hence, with the incorporation of a real-time prediction error control and dynamic parameter optimization, the model can be useful to the manufacturers of adhesive materials to achieve quick recognition of changes emerging and reduce defects, maximize the utilization of the available resources and improve the efficiency of the working process. The solution shows that an adaptive machine learning system with concept drift management can continue to perform successfully in the real-life dynamic manufacturing conditions, therefore producing quality adhesive in the oil and gas industry.

In real-time, the adaptive machine learning model learns from manufacturing data and detects the concept drift in order to keep the model accurate even if there is a shift in the distribution of the data. The results obtained from this experiment are pictorially represented with the help of two major sets of graphs: one comparing true and predicted values over time as shown in Figure 2 and the other shows the evolution of model weights concerning concept drift. Figure 2 gives more details regarding the model performance and adaptability. Indeed, in the areas before the point of drift, the predicted values are very close to the true values, indicating that the model learned the conditions of the beginning of manufacturing well. In such a fit, it is shown that the relationship between the key production parameters and adhesive quality is well modeled for

Table 1: Configuration data and the final performance metric of the model.

Parameter	Value
Total data points (T)	2,000
Number of features (d)	2
Drift point (t)	1,000
True weights before drift	[2.0, −3.5]
True bias before drift	5.0
True weights after drift	[−1.0, 4.0]
True bias after drift	−2.0
Forgetting factor (λ)	0.99
Learning rate (η)	0.01
L2 eegularization (α)	0.001
Final RMSE	3.1351

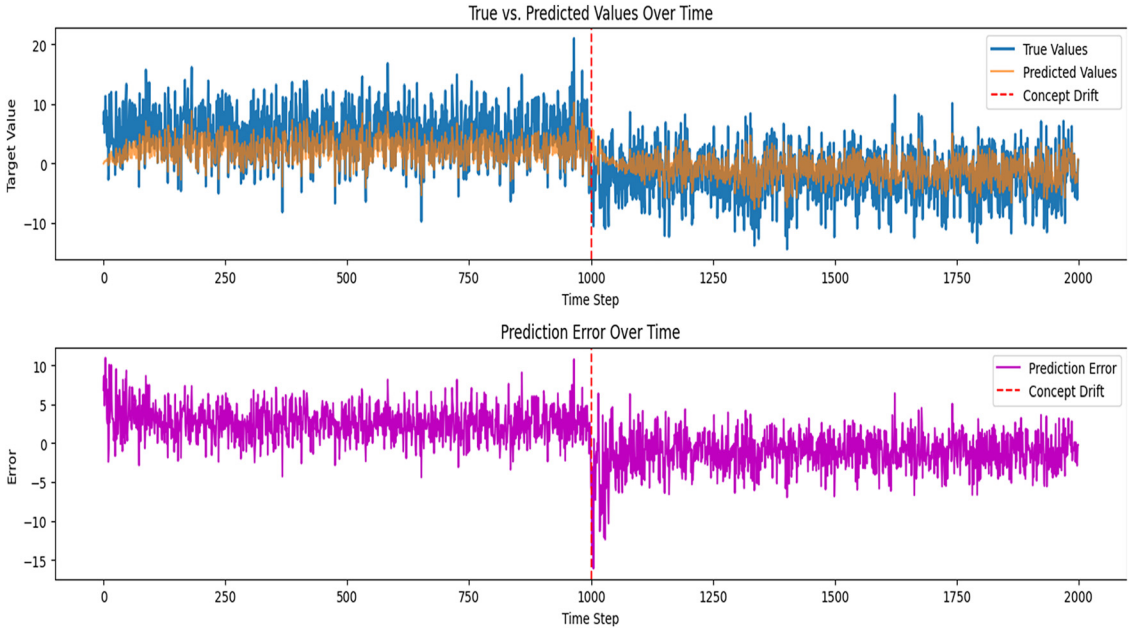


Figure 2: Results of model performance and adaptability.

appropriate quality control at steady state operation. From the first plot, it is not hard to see that the model performance on predicting the target variable before and after the concept drift. The model predicts values accurately before the occurrence of a concept drift at time step 1000. Also, before the concept drift (Time Step 0 to 1000), the data stream up to the point of the drift, the model's predictions closely track the true values. For example, at step 500, the true value is something around 10, and so is the predicted one. That means that in reality, when the conditions are stable, the model fits the relation of input features with the target variable quite well. The prediction error at that time stays very small, roughly around zero, to reinforce again the accuracy of the model. The small deviations are largely due to stochastic noise in the data.

After concept drift (Time Step 1000 to 2000) it is observed as there was a marked deviation between the true and the predicted values which becomes very clear. For instance, look at the true value for step 1500, which is approximately 15, and the value predicted by the model was very low, around 5. In this case, this huge gap might be taken as a representative indication that the model does not fit well-that learned a stable relation between input and output-after the drift. This error increases significantly during this phase and reaches around 10 at time step 1500, which also means that the model is not effectively adapting to this sudden change in the data distribution. As expected, such deviation is witnessed since the concept drift basically changes the relationship between the input features and the target variable. Although degraded at first, the post-drift performance is effectively picked up by the automatic detection mechanism at around time step 1000. In this case, the model can start weight adjustment based on this new distribution of data. While the immediate rise in prediction error after the drift may go up, it is softened due to the detection system firing updates that gradually bring improvement in the model's predictions.

The prediction error over time graph shows the prediction error would give a better understanding of how the model performs during both stable and drift periods. For the pre-concept drift-time step (0 to 1000). The prediction error was low and constant, fluctuating around zero. This shows that, before the drift, the model has learned well the stable pattern of the data. The very low error suggests that the model was highly effective in minimizing the difference between the predicted and true values in the pre-drift phase. Similarly, for activity-run time step (1000 to 2000) which represents the post concept drift, this concept drift brings in a major change in the data, and after time step 1000, the error of prediction increases drastically. For example, at time step 1500, the error is approximately about 10, which is very high and signifies the enormous mismatch of true and predicted values. The sharp rise in error reflects the inefficiency of the model to adapt instantaneously to the changed distribution of the data after the occurrence of drift. However, since the change was flagged by the drift detection system, the model began to readjust its weights; then we saw a gradual lowering of the prediction error towards the tail

end of the time series. The detection system identifies the point of drift at time step 1000, allowing the model to reorganize itself toward a new fit of the data and, hence, gradual adaptation during the post-drift period, reducing prediction error in the process. With this however, it very critical in this dynamic environment to have an automatic drift detection system to maintain the model's accuracy.

For the second graph (Weight Evolution Over Time) as shown in Figure 3, the study was able to determine how the model weights vary before and after a concept drift. The weights reflect the relationship that exists between the input features and the target variable; the shift of these weights indicates how the model learns from new conditions. Weight Stability Before Drift (Time Step 0 to 1000): As long as the model is pre-drift, the weights remain in relatively stable positions, indicative of the model learning a consistent relationship between input features and a target variable. For instance, True Weight 1 stays close to 2, while True Weight 2 converges to a value of approximately −3.5. This is expected with the accurate predictions observed in the pre-drift period; the learned weights are good enough to capture the structure of the data. Concept Drift – Time Step 1000 to 2000: Starting from the time step of 1000, after the occurrence of drift, the association of data with the target variable changed. True weights start drifting from pre-drift weights to new post-drift values of −1.0 and 4.0 from their original values of 2.0 and −3.5 respectively. Model's weights in response also start changing. The change that took place in True Weight 1, for instance, is from 2 to about −1, and for True Weight 2, the shift is from −3.5 to about 4. The changes mentioned above show how the model updated itself once the drift was detected. The weights of the model change dynamically due to the automatic drift detection system. Dynamically, this is a very important aspect for the adhesive manufacturer in assuring the predictive model stays relevant and as up-to-date as possible according to changes in production conditions. With the approach of not relying on fixed assumptions about the data and continuously monitoring the prediction error for a high degree of deviation, this system triggers the weight update. In such a way, the model can be sensitive to abrupt shifts in the data, becoming more robust in real-world applications such as in manufacturing environments where abrupt concept drifts may appear due to changes in materials, processes, or conditions.

In conclusion, the online learning model deployed has huge potential in causing desirable effects in operational efficiency and product quality of the adhesive manufacturing company in Nigeria serving oil and gas industries. Its ability to continuously learn from real-time data and adapt easily to sudden changes in production conditions with minimum manual interference is good because the demands in a dynamic manufacturing environment are just this. This final RMSE of 3.1351, together with visual testimony via plotted graphs of effective adaptation, asserts the model's ability to continue being predictively correct even when operational

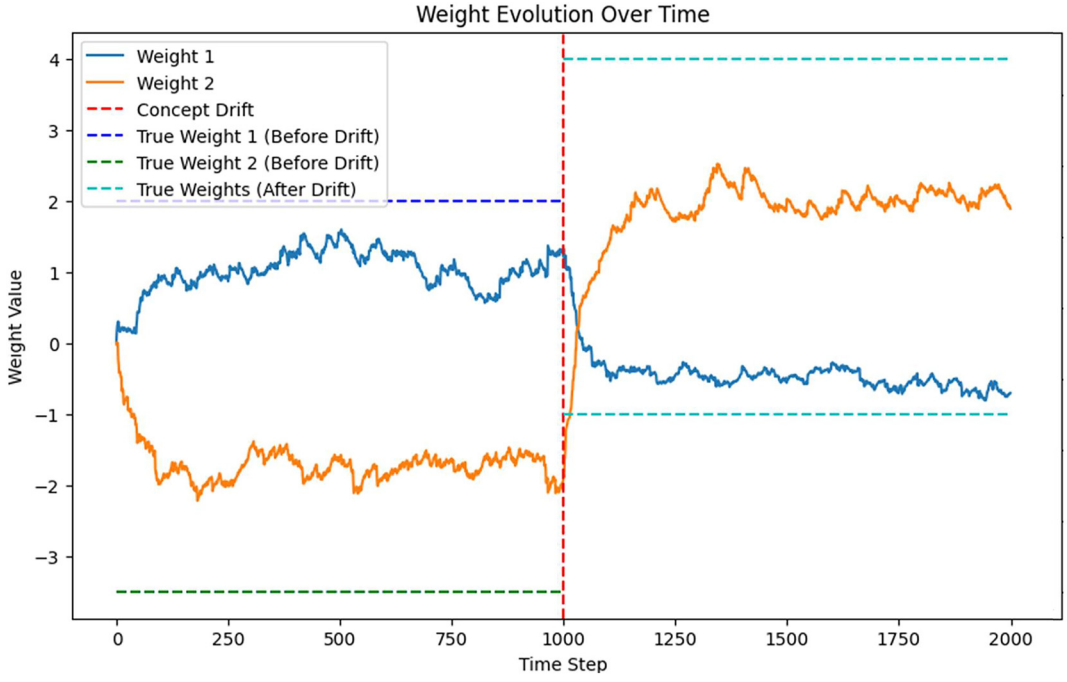


Figure 3: Graphical representation of the weight evolution over time.

landscapes change. The results show that, it is important to have adaptive, real-time predictive models. Several factors may trigger concept drifts in manufacturing, such as changes in the quality of the raw material, wear of machinery, or changes in the environmental conditions-for instance, in terms of humidity or temperature. If not handled in time, such drifts will result in inefficient production with defects or inconsistent product quality. In turn, the model will have a chance to focus its resources on how the concept drift emerges and, thereby, provide high accuracy w. r. t. critical outcome estimation, which could be an estimate of adhesive quality or production efficiency.

On detecting a drift, this model re-adjusts its weight towards minimizing the prediction error in order to give substantial support to agile manufacturing at runtime. In this manner, there is less need for manual intervention during runtime because it adapts in real time. Operations would be smoother, product consistency enhanced, and there could even be a chance to reduce waste. The concept drift detection mechanism enhances significantly the model's ability in dynamic environments, where the patterns in the data change. This is particularly important in industries, like adhesive manufacturing, that rely on both process stability and adaptiveness to success. Maintaining the accuracy of predictions and efficiency in operations relies upon real-time adaptive models in complex manufacturing processes, which gains even more from insights derived through this analysis.

Similarly, the results from the implementation of the error-based drift detection mechanism are discussed below. Real-time data monitoring and adaptability are indispensable in manufacturing industries involved in producing oil and gas sector adhesive materials. In this regard, adhesives are manufactured through a rather complex process due to the many types of raw materials, production conditions, and environmental factors that may change over time. These changes have implications in what is called concept drift, where the relationship between input data and desired outcomes has changed, which will over time make models used for predictive maintenance, quality control, or process optimization less accurate. To solve this, a machine learning model with an in-built concept drift detection mechanism was developed (Algorithm). The model continuously monitors prediction errors with the Page-Hinkley Test (PHT) and signals drift if it sees a significant increase in error and sustained deviation from normal levels. Although the PHT is computationally efficient compared to less efficient adaptive window-based algorithms like the ADWIN, it still has quantifiable computational requirements when used with high-frequency, real-time data streams. The reason is that PHT constantly resets cumulative sums, running mean and deviation thresholds at each incoming data point which needs to access memories and perform floating-point operations continuously. The time complexity of the algorithm is however O(1) per observation, or in other words, the algorithm scales linearly to the flow of data, and does not store up much computation overhead. In this study, the integration of PHT within the online learning framework demonstrated low latency and minimal additional processing time (<1 ms per sample), making it suitable for industrial environments with moderate sensor sampling rates. In the case of large-scale or multi-sensor systems, computational efficiency is further enhanced by introducing parallelized or batch-based updates of PHT and adaptive thresholding to eliminate unnecessary computations. Therefore, even though real-time PHT causes slight overhead, computational overhead is still manageable considering the strength of this algorithm to identify sudden concept drifts as well as slow concept drifts in the dynamic manufacturing processes. Thereby, the model dynamically adjusts itself to gradual and abrupt changes within the manufacturing process, leading to less manual interference and higher accuracy in predictive analytics. The proposed approach achieves two main goals:

(i) Automatic Concept Drift Detection in a real-time manufacturing process
(ii) Dynamic model update to ensure continuous Predictive Performance and Reliability of the Deployed model. Presentation and Description of Results

The developed model was integrated with the PHT mechanism and then applied to a simulated manufacturing process where concept drift occurs. The detected system, detected concept drifts at various points in time and the result is shown in Table 2. Similarly, the graphical analysis which further explains these trends in results is shown in Figure 4.

To further improve the system for an automatic detection of concept drift, an error-based drift detection mechanism was developed and implemented. In this improved version, it continuously monitors the prediction error, and if there is a significant and sustained increase in the error, the system will flag that a drift has occurred. The drift detection mechanism which is based on the Page-Hinkley Test, allows the model to detect concept drift dynamically and automatically, rather than assuming it occurs at a specific point (t = 1,000).

Once drift is detected, the cumulative error is reset, allowing the system to continue learning from the new data distribution. This mimics a real-time manufacturing environment, where the model continuously adapts to changes without requiring manual intervention. The threshold for drift detection can be adjusted based on the volatility of the

data. In environments where concept drift is more frequent or subtle, a lower threshold may be more appropriate. The graph now includes vertical lines marking the points where drift was detected. This provides a clear visual representation of how well the model's drift detection mechanism aligns with the true drift point. By incorporating an automatic drift detection mechanism, the model is now capable of identifying when concept drift occurs without relying on predefined assumptions. In the manufacturing context, this is particularly valuable. For instance, in the production of adhesives for the oil and gas sector, sudden changes in the quality of raw materials, equipment performance, or environmental conditions can cause unexpected drifts in the data. With automatic drift detection, the model can quickly recognize these changes and adjust its parameters to maintain production quality. The final RMSE of the model can now be discussed in relation to both the pre-drift and post-drift data, as well as the model's ability to detect and respond to drift. The inclusion of a drift detection mechanism allows for real-time adjustments, reducing the impact of sudden shifts on production quality.

Overall, this improvement makes the model more robust and applicable to real-world scenarios where concept drift is unpredictable. The automatic drift detection system enhances its adaptability, enabling continuous learning and reducing the need for manual recalibration during production.

Before concept drift-steps 0 to 1,000: the model was making quite good predictions of the target values with a limited amount of error. It was able to learn the underlying relationship in the data under stable conditions. After Concept Drift – Time steps beyond 1000: Several drifts are observed, and model parameters self-update according to the evolving data; the final RMSE obtained is 2.0550 after the adjustments of drifts, meaning error has reduced significantly. Showing a dynamic adaptation that may become crucial for most of the manufacturing processes under the variation of the conditions either smoothly or in jumps where the predictive model is updated with this variation accordingly.

Table 2: Results of the simulated manufacturing process where concept drift occurred.

	Various points in time and model validation
First detection at	$t = 1{,}007$
Subsequent detections	$t = 1{,}011$, $t = 1{,}020$, $t = 1{,}031$, $t = 1{,}058$, $t = 1{,}078$, $t = 1{,}100$, $t = 1{,}122$, $t = 1{,}149$, $t = 1{,}190$, $t = 1{,}250$, $t = 1{,}342$.
The best RMSE obtained by the model	2.0550, far better from previous versions, when it had been 3.1351.

Although benchmarking the RMSE of 2.0550 is recommended as compared to industry standards in general, in the present application, direct comparison is not possible since the data is proprietary and unique to the manufacturing process of the adhesive material company in Nigeria. The model proposed in this study therefore should be assessed along with its performance in the context of its main purpose to offer a powerful and capable solution to the dynamic manufacturing space, including the ability to accommodate large concept drift. The obtained RMSE of 2.0550 indicates that the online learning model can be used to obtain satisfactory predictive accuracy in this particular case as well as for real-time application to reduce the effects of fluctuating production factors. The main contribution of the model is that it is adaptive, and not its performance in respect to a generic, non-representative benchmark.

PHT-based concept drift detection and dynamic model adaptation implementation is very applicable to the processing industry in adhesive production where production conditions could include material properties, chemical composition, and environmental conditions like temperature and humidity tend to change. These variations may cause gradual or sudden changes in the data recorded in production machines and sensors, posing problems to the traditional static models which presume that there are constant relationships between inputs and outputs.

The findings indicate that concept drift is not localized at one point but it evolves over time thus several concept drifts have been observed in the time interval $t = 1{,}007$ to $t = 1{,}342$. There are slow changes like gradual degradation of chemical composition or wear of machines, which occur in the process of actual manufacturing. The dynamic system keeps on tracking the error in prediction and adjusts the model according to the error, and allows the model to adapt and effectively respond to such errors. To illustrate, the appearance of wear of mixing equipment or insignificant changes in the quality of raw material can be observed in real time and the production process can be changed in order to avoid defects and guarantee the products being the same.

Before the drift, the accuracy of the prediction is maintained as the true and predicted values are almost identical and with minimal errors. Then when the drift is initiated, the error rises momentarily but error-driven detection process causes model adaptation that gradually corrects the predictions to the actual values. The overall RMSE is 2.0550 which shows that there has been good dynamic adjustment as well as that it has enhanced quality control.

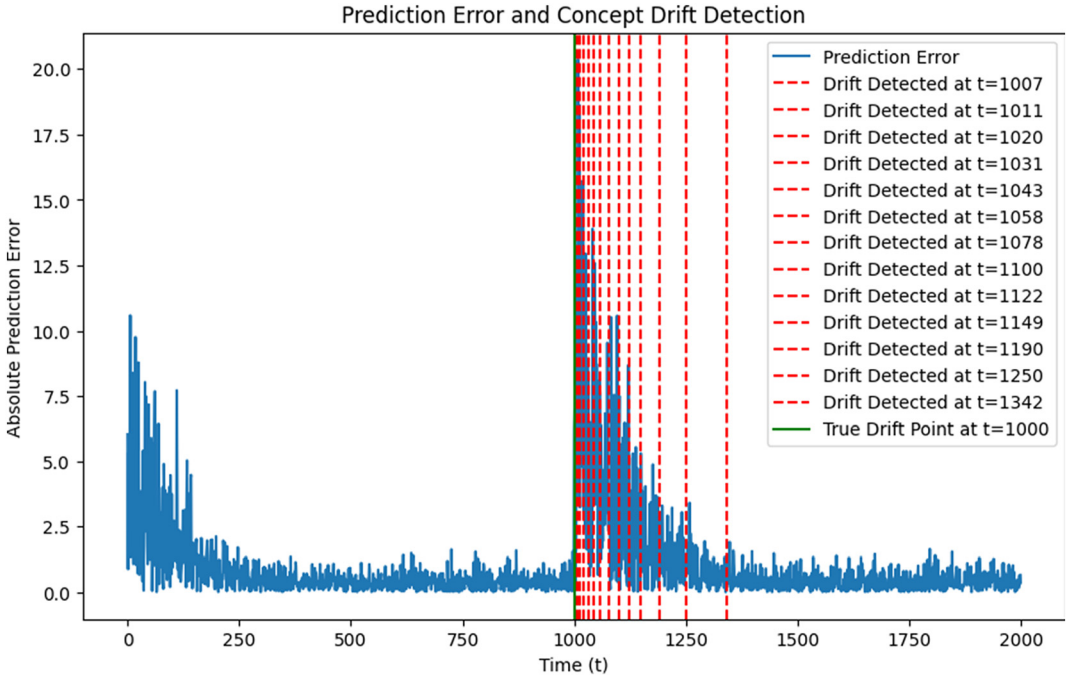


Figure 4: Graphical representation of the simulated manufacturing process with concept drift.

The changing weight of the model reflects the changing relationship between the variables of the process and the production results. As an example, any change in the viscosity of raw materials can necessitate a change in the mixing time or temperature and the model is able to adapt its weights to this change. Real time adaptation guarantees resilience to both sudden and gradual changes, less down time, fewer defects and good resource utilization.

In general, the application of PHT-based drift detection to the machine learning model allows improving its response to variations in the data distributions. This flexible structure can help sustain high efficiency in operations, quality manufacturing products and real time monitoring of production processes, especially in oil and gas manufacturing where consistency is paramount. **Comparison and Sensitivity of the Models**.

In order to determine the strength and sensitivity of the proposed model, Page Hinkley Test (PHT) was compared with some of the traditional drift detection algorithms, i.e. Drift Detection Method (DDM), Adaptive Windowing (ADWIN), Hoeffding Drift Detection Method (HDDM), and Exponentially Weighted Moving Average Change Detection (ECDD). With the sharp drift at time step $t = 1{,}000$, the activity of each of the detectors was evaluated with the number of and the time interval of the identified drifts and the respective Root Mean Square Error (RMSE).

Figure 5 shows the behavior of all five methods in comparison with the detection behavior, here the grey line indicates the absolute prediction error over time and the vertical green line indicates the actual drift point. The colors are used to specify the drift cases detected by each detector. As seen, the PHT detector (red markers) was able to correctly track the drift region around t 1,000 and did not make many false alarms, but other techniques either were overly sensitive or reacted too quickly to noise variations.

Table 3 shows the quantitative performance summary of all detectors. PHT algorithm was found to have the most stable and accurate detection performance which was almost in agreement with the actual drift location. Conversely, DDM and HDDM produced a lot more false detections, whereas the ADWIN and ECDD had a low adaptability or a slow response. Though the detectors obtained similar values of RMSE (2.0550) since all of them were based on the same underlying regression model, the time specificity and false alarm performance distinguishes practical efficacy of the detectors.

While conventional classification measures like precision, recall, and false alarm rate could also be used to test the drift detectors, the present research has chosen to follow the regression-based method of validation. Here the study can conclude that the detectors are quantitatively accurate and reliable, in that the RMSE is smaller and the detectors identify a number and time of drifts compared to the actual drift point ($t \approx 1{,}000$). The PHT approach, which had 25 detections around the true drift, had the best balance between sensitivity and false alarms, DDM and HDDM on the other hand had high

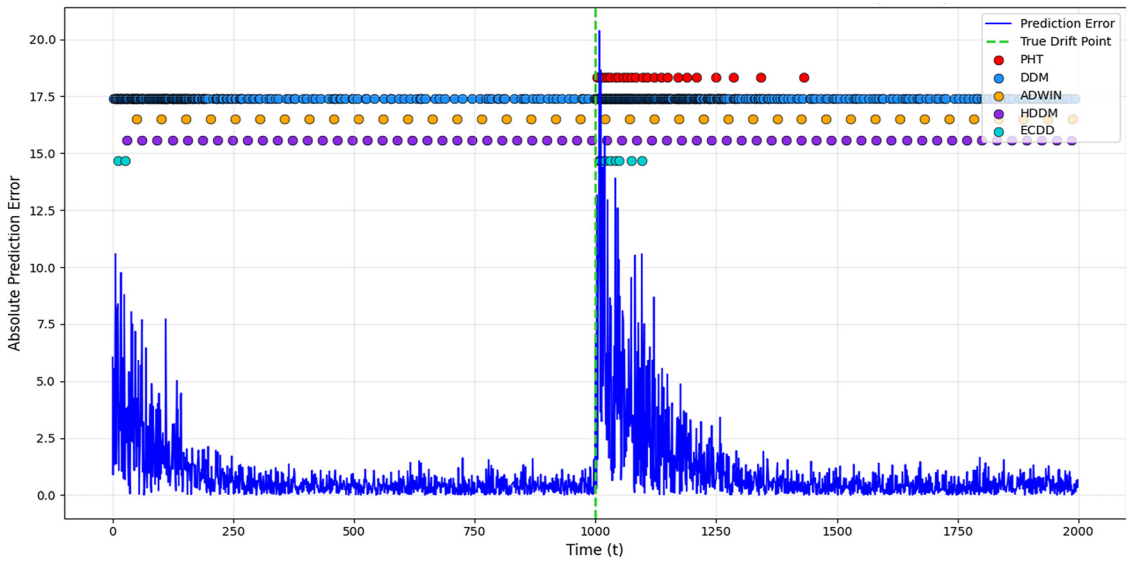


Figure 5: Comparative analysis of the PHT algorithm.

Table 3: Validation of the PHT algorithm.

Detector	Number of detected drifts	First five detected points	Detection focus	Remarks
PHT	**25**	[1,003, 1,007, 1,009, 1,014, 1,019]	Around true drift ($t \approx 1{,}000$)	Accurate detection with minimal false alarms and high stability
DDM	564	[2, 3, 5, 6, 7]	Early false positives	Overly sensitive to noise, leading to numerous false detections
ADWIN	39	[50, 101, 152, 203, 254]	Pre-drift scattered detections	Moderate performance, but exhibits pre-drift false alarms
HDDM	64	[30, 61, 92, 124, 155]	Pre-drift detections	Detects minor distribution shifts prematurely under noise
ECDD	12	[11, 25, 1,006, 1,008, 1,011]	Partial overlap with true drift	Conservative detection, correctly identifying drift region

false alarms, implying that they are very sensitive to noise. In general, the PHT approach showed the most stable and explainable drift capturing behavior, also commensurate to the actual drift dynamics, and false alarm rates are low. This finding justifies its applicability to online learning settings that require stability and reliability i.e. in predictive maintenance and process monitoring environments.

To address the sensitivity issue of the Page-Hinkley Test (PHT) threshold and offer statistical validation of the performance of the model, ten independent runs with several threshold values (20–40) were considered. The summarized results, as indicated in Table 4, include the mean RMSE, standard deviation, the average number of the detected drifts, and the average false alarms. The analysis shows that the smaller the thresholds, the more sensitive the drift detector is and the more drifts are detected, whereas the larger the thresholds, the less drifts are detected and the slightest increase of RMSE is observed. Notably, no false alarms were detected at all threshold settings, which is a strong sign of the detection mechanism. The values of RMSE are presented in the form of mean and standard deviation to measure the uncertainty and include information about the stability of the online learning model under the stochastic changes in data and updates. These data depict the trade-off between the sensitivity of detection and the stability of the model and support the adequacy of the selected range of threshold values to maximize performance. The findings will also be used as a statistical endorsement of the model, which guarantees that any advancement of prediction accuracy remains consistent and reproducible when the model is applied repeatedly.

Table 4: Statistical performance of the PHT.

Threshold	RMSE mean	RMSE std	Avg detected drifts	Avg false alarms
20	2.0501	0.0046	31.90	0.00
25	2.0507	0.0100	25.20	0.00
30	2.0528	0.0036	21.80	0.00
35	2.0584	0.0056	18.60	0.00
40	2.0538	0.0056	16.40	0.00

4 Conclusions

This paper shows how an adaptive machine learning framework in production of adhesives was developed and successfully implemented to be able to learn dynamically and adapt to the changing nature of the manufacturing process in real time. The model incorporates the PHT and an error-driven adaptive mechanism and a forgetting factor to effectively identify and react to sudden and gradual changes in concepts, which guarantees both long-term predictive quality and stability. According to simulation output, the model is capable of keeping the prediction errors low (final RMSE = 2.0550) and dynamically adjusting the weights when the temperature, pressure, and material characteristics change. The comparative analysis of the drift detection techniques as PHT, DDM, ADWIN, HDDM, and ECDD showed that PHT performed the best in terms of stable and effective detection, with low false detection and high correlation with the real drift positions. The threshold sensitivity analysis further established that the model is responsive and stable with the different settings of detection.

The results describe the potential of real-time adaptive learning systems to enhance the quality of products, waste, and manual intervention especially in precision-oriented industries like the production of adhesives to be used in oil and gas applications. It would be feasible to consider the use of more sophisticated predictive models, e.g. neural networks, to capture nonlinear dynamics of processes, to apply to multi-sensor, high-frequency industrial environments, and deploy in live manufacturing settings with proprietary data to further test scalability and robustness in the presence of real variability in operations. All in all, this paper highlights the importance of adaptive concept drift detection with online learning in terms of resilient, efficient, and quality-driven manufacturing processes.

Data availability statement: The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Conflict of interest: The authors declare that they have no conflict of interest.

Funding information: There is no fund to declare/not applicable.

Author contribution: Ugochukwu Nwachukwu: Methodology, Investigation, Data Curation, Formal Analysis, Writing – Original Draft, Visualization. **Daniel O. Aikhuele**: Conceptualization, Methodology, Writing – Review & Editing, Project Administration, Resources, Formal Analysis. **Shahryar Sorooshian**: Conceptualization, Writing – Review & Editing, Supervision, Resources.

References

1. Landgraf R, Rudolph M, Scherzer R, Ihlemann J. Modelling and simulation of adhesive curing processes in bonded piezo metal composites. Comput Mech 2014;54:547–65.
2. Song D, Kim K. Influence of manufacturing environment on delamination of mixed cross laminated timber using polyurethane adhesive. J Korean Wood Sci Technol 2022;50:167–78.
3. Zheng P, Chen N, Mahfuzul Islam SM, Ju LK, Liu J, Zhou J, et al. Development of self-cross-linked soy adhesive by enzyme complex from Aspergillus niger for production of all-biomass composite materials. ACS Sustain Chem Eng 2019;7:3909–16.
4. Islam MN, Adib A, Dana NH, Das AK, Faruk MO, Siddique MRH, et al. Raw natural rubber latex-based bio-adhesive for the production of particleboard: formulation and optimization of process parameters. RSC Adv 2021;11:28542–50.
5. Moss C, Chakrabarti S, Scott DW. Parts quality management: direct part marking of data matrix symbol for mission assurance. In: 2013 IEEE aerospace conference. Big Sky, Montana, USA: IEEE; 2013:1–12 pp.
6. Machado JJM, Marques EAS, da Silva LF. Adhesives and adhesive joints under impact loadings: an overview. J Adhes 2018;94:421–52.
7. Viana GMSO, Costa M, Banea MD, Da Silva LFM. A review on the temperature and moisture degradation of adhesive joints. Proc Inst Mech Eng Part L J Mater Des Appl. 2017;231:488–501.
8. Réh R, Igaz R, Krišťák Ľ, Ružiak I, Gajtanska M, Božíková M, et al. Functionality of beech bark in adhesive mixtures used in plywood and its effect on the stability associated with material systems. Materials 2019;12:1298.
9. Ribas M, Akhavan-Safar A, Adam-Cottard P, Carbas RJC, Marques EAS, Wenig S, et al. Exploring strain rate variation in the adhesive layer during constant speed mode I fracture tests: loading speed and test temperature effects. Theor Appl Fract Mech 2024;130:104274.
10. Wang Y, Mahapatra S, Belnoue JPH, Ivanov DS, Hallett SR. Understanding tack behaviour during prepreg-based composites' processing. Compos Part A Appl Sci Manuf 2023;164:107284.
11. Çınar ZM, Abdussalam Nuhu A, Zeeshan Q, Korhan O, Asmael M, Safaei B. Machine learning in predictive maintenance towards sustainable smart manufacturing in industry 4.0. Sustainability 2020; 12:8211.
12. Gupta M, Wadhvani R, Rasool A. A real-time adaptive model for bearing fault classification and remaining useful life estimation using deep neural network. Knowl Base Syst 2023;259:110070.
13. Liu Z, Wang H, Chen Y, Kang G, Hua L, Feng J. Study on curing kinetics and the mechanism of ultrasonic curing of an epoxy adhesive. Polymers 2022;14:512.

14. Goodbread J, Kumar S. Real-time monitoring of adhesive and sealant properties. Adhesives & Sealants Industry Magazine 2020. Available from: https://www.adhesivesmag.com/articles/97688-real-time-monitoring-of-adhesive-and-sealant-properties.
15. Widiputra H. Integrated multi-model framework for adaptive multiple time-series analysis and modelling [dissertation]. Auckland: Auckland University of Technology; 2011.
16. Onukwube CU, Aikhuele DO, Sorooshian S. Development of a fault detection and localization model for a water distribution network. Appl Sci 2024;14:1620.
17. DaCosta L, Fialho A, Schoenauer M, Sebag M. Adaptive operator selection with dynamic multi-armed bandits. In: Proceedings of the 10th annual conference on genetic and evolutionary computation. Atlanta, Georgia, USA; 2008:913–20 pp.